

Investigating the Role of AI in Emergency Management: Use Cases, Challenges, and Opportunities

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Though natural disasters and other hazards remain a persistent threat, Emergency Managers (EMs) face substantial barriers to effectively completing their life-saving job functions. Pervasive staffing challenges and diminishing funding sources complicate a profession tasked with timely, critical decision-making under uncertainty. While technology presents a significant opportunity to complement EMs' workflows, commercial products are extremely costly and typically serve narrow use cases. To understand how EMs use and envision using technology, and in particular AI, we conducted in-depth interviews with 32 professionals spanning different geographic regions within the US. Through iterative thematic analysis and affinity mapping, we identified 7 key categories of AI technology use cases for EMs' core job functions, representing 136 distinct specific use cases. In the context of these use cases, we describe the limitations of current technologies, provide guidance on EMs' design needs, and highlight opportunities for computing research to support them.

CCS Concepts: • **Human-centered computing** → **User studies**; *Empirical studies in interaction design*; • **Applied computing** → **Computing in government**.

Additional Key Words and Phrases: Emergency Management; Crisis Informatics; Artificial Intelligence; Qualitative Methods; Domain Expert Perspectives; Public Sector Technology; Disaster Management

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1 Introduction

In 2024 alone, natural disasters inflicted over \$182.7 billion in damages across the United States—a figure projected to grow as climate change intensifies the frequency and severity of extreme weather events [Smith 2025]. Emergency managers (EMs) coordinate the complex, life-saving work of preparing for and responding to all disasters—natural and man-made—in their communities, known as *emergency management* (EM). U.S. emergency

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Fig. 1. The 4 Phases of Emergency Management: mitigation, preparedness, response, and recovery.

management is structured around a cyclical four-phase model—mitigation, preparedness, response, and recovery (Figure 1)—that organizes the work EMs do before, during, and after a disaster.

Central to U.S. EMs’ practice is the *Whole Community* approach [Federal Emergency Management Agency 2011]: genuinely engaging community members in the 4 phases of EM, especially those most vulnerable to disaster impacts. This doctrine emerged from lessons learned from historical disparities in disaster outcomes, such as Hurricane Katrina in 2005, where failures in EM disproportionately harmed low-income and BIPOC communities [FEMA Current and Former Workers 2025]. Despite the explicit focus of EM on marginalized populations and the universal relevance of disaster preparedness, the field is remarkably under-studied by the FAccT, HCI, and broader computing research community. This gap is particularly concerning given the growing popularity of Artificial Intelligence (AI) applications in the field: federal agencies are exploring AI for disaster response [Agency 2022; Federal Emergency Management Agency 2025; U.S. Department of Homeland Security 2024], professional conferences feature workshops on AI integration in disasters [International Association of Emergency Managers 2025], and commercial vendors market AI-enhanced software [esri 2026; Juvare 2026]. Throughout this paper, we use the definition of “AI system” in the EU AI Act: a “machine-based system designed to operate with varying levels of autonomy that infers, from its inputs, how to generate outputs such as predictions, content, recommendations, or decisions that can influence physical or virtual environments” [European Parliament and Council of the European Union 2024]. It is unclear whether the rapid development of these technologies involves systematic engagement with the EMs who would actually use these systems. We aim to understand the most pressing of these constraints, how practitioners navigate them, and where AI may genuinely help.

The literature contains gaps relevant to responsible AI system design, deployment, and auditing. Training datasets exhibit geographic and linguistic biases: damage assessment models show reduced performance on Global South contexts, crisis NLP datasets are predominantly English-centric, and search-and-rescue datasets are geographically limited to European terrain [Alam et al. 2021b; Božić-Štulić et al. 2019; Gupta et al. 2019]. Explainability and transparency remain underexplored in EM, with a recent systematic review identifying explainable AI methods for disaster decision support as a key research frontier [Ghaffarian et al. 2023]. Perhaps most critically, a concerning amount of AI for EM publications do not report substantive engagement with EMs—the intended end users of these technologies [Moitra et al. 2022; Purohit et al. 2025]. While crisis informatics has examined social media use during disasters, and human-AI collaboration reviews identify patterns for decision support and trust calibration [Domfeh and Dancy 2025; Reuter and Kaufhold 2018], empirical research directly investigating professional EMs’ perspectives on AI remains limited. This gap risks developing AI systems that fail to address practitioners’ actual operational needs or integrate effectively with existing workflows.

Our study addresses this gap by centering the perspectives of 32 U.S.-based EMs on technology use and AI potential in their profession. Rather than beginning with technological capabilities and searching for applications, we began with practitioners: their daily workflows, their frustrations with existing tools, their experiences with emerging AI technologies, and their perspectives on where AI might—and might not—prove useful.

Our research addresses three primary questions:

- **RQ1:** How do emergency managers currently use technology—including AI—in their workflows, and what are the limitations and successes of these technologies?
- **RQ2:** What are emergency managers' perspectives on the usefulness of AI in their work context, including both opportunities and concerns?
- **RQ3:** For which use cases do emergency managers imagine that AI tools would be the most helpful and how feasible are their desires given the current state of each use case?

The remainder of this paper proceeds as follows. We first review relevant background on EM, technology adoption in the profession, and related computing research. We then describe our interview methodology, including participant recruitment, protocol development, and analytical approach. Our findings section presents contextual factors shaping technology adoption followed by detailed analysis of seven functional use case categories. We conclude with a discussion of implications for AI research and design, opportunities for near-term impact, and limitations of our study.

2 Background and Related Work

EM remains notably absent from studies of domain expert perspectives on AI. One of the only recent studies centered on the needs of disaster risk practitioners included a population primarily of scientists and academics rather than operational EMs [Moitra et al. 2022]. Our study addresses this gap by centering the perspectives of frontline EMs who would ultimately use AI tools in operational settings, contributing to a more complete understanding of expert-AI interaction across high-stakes public sector contexts. The remainder of this section covers important context about EM in the US, a brief history of relevant technology in the field, and the current landscape of AI in crisis and disaster management.

2.1 Emergency Management: Doctrine and Theory

U.S. EM is structured around the four-phase model—mitigation, preparedness, response, and recovery (Figure 1)—codified in a foundational 1979 National Governors Association report [Baird 2010; National Governors Association 1979]. This cyclical model was accompanied by a paradigm shift toward the *all-hazards approach*, which moved the profession away from hazard-specific planning toward capabilities-based preparedness that addresses multiple threat types through common operational frameworks [Bodas et al. 2012; Federal Emergency Management Agency 1996]. Operationally, EMs coordinate through the National Incident Management System (NIMS) and Incident Command System (ICS), standardized frameworks mandated by presidential directive following September 11th that structure how personnel, resources, and information flow during incidents [Bigley and Roberts 2001; The White House 2003; U.S. Department of Homeland Security and Federal Emergency Management Agency 2017]. These frameworks emphasize flexibility and scalability, enabling the same organizational structures to manage events ranging from localized flooding to multi-state disasters [Jensen and Thompson 2016]. The professionalization of EM has progressed substantially since its civil defense origins, with the field now supported by established competency frameworks, professional certifications, and accreditation standards [Drabek 1987; Emergency Management Accreditation Program 2022; Feldmann-Jensen et al. 2019; International Association of Emergency Managers and FEMA Emergency Management Higher Education Project 2007].

Executing services within this model, the EM profession faces persistent structural challenges that shape how practitioners engage with technology. A comprehensive 2025 national study found that 57% of local EM

agencies operate with one or fewer full-time employees, 56% of directors hold additional official duties beyond their EM role (“dual-hatting”), and 15% of local director positions are unpaid volunteers [Argonne National Laboratory and National Preparedness Analytics Center 2025]. These staffing constraints are compounded by chronic underfunding, with state EM budgets ranging from \$500,000 to \$530 million and median budgets under \$10 million [Deloitte Center for Government Insights and National Emergency Management Association 2025; National Emergency Management Association 2024]. The resulting workload has documented consequences: studies report burnout prevalence exceeding 50% among emergency services personnel, with staff shortages directly linked to employee attrition [Patel et al. 2023; Phillips 2018; U.S. Government Accountability Office 2023]. These workforce realities have created an environment where EMs often view technology adoption as a tool for extending limited human capacity rather than a credible avenue for job automation and/or replacement [Tapia and Moore 2014].

2.2 Technology Adoption and Use in Emergency Management

Research on EMs’ technology adoption has revealed distinct patterns. Adapting the Unified Theory of Acceptance and Use of Technology (UTAUT) to examine decision support software for EMs, Jennings et al. found that social influence, inter-agency collaboration, and community social vulnerability significantly shape local EMs’ intent to adopt technology. Similarly, surveys of state EMs have consistently identified financial constraints and insufficient support from elected officials as perennial barriers to technology adoption, with higher-performing states more likely to leverage technology across all phases of EM [Reddick 2011]. These findings suggest that technology adoption in EM—especially in the public sector—is tightly coupled with agencies’ organizational, political, and resource constraints.

Despite the variety of contemporary EM software—a 2022 market survey identified 38 commercial incident management software products with varying capabilities for GIS integration, interoperability, and alert management [DHS Science and Technology Directorate and National Urban Security Technology Laboratory 2022]—our interview participants overwhelmingly described using two main EM-specific software applications: WebEOC and GIS. Emergency Operations Centers (EOCs) rely on platforms like WebEOC [Juvare, LLC 2025] for shared request tracking and situational awareness during large-scale events [Federal Emergency Management Agency 2019]. Geographic information systems (GIS) have achieved widespread adoption with over 90% of state EM agencies employing dedicated GIS staff by 2011 [Council 2007; Tomaszewski et al. 2015]. Interoperability remains a persistent challenge: the nation’s approximately 6,000 911 call centers each manage their own technology systems, and incompatibilities in communication equipment have hindered coordinated response efforts [DHS Science and Technology Directorate 2024; Munkvold 2023; U.S. Government Accountability Office 2018].

For almost 2 decades, the field of crisis informatics has examined how affected populations and digital volunteers use technology [Danielsson et al. 2014; Endsley 1995; Palen and Anderson 2016; Reuter and Kaufhold 2018; Vieweg et al. 2010], with comparatively less attention to the perspectives and workflows of professional EMs themselves [Hiltz et al. 2014; Reuter et al. 2018]. Studies with direct practitioner engagement note organizational barriers to social media adoption and a pragmatic orientation towards “good enough” solutions given resource constraints [Tapia and Moore 2014]. As EM agencies express growing interest in advanced tools including AI—while acknowledging commercial offerings are “not quite there yet” [Deloitte Center for Government Insights and National Emergency Management Association 2025]—there is a need for empirical research that centers practitioner perspectives on how emerging technologies might integrate with their existing workflows and address their actual operational challenges.

2.3 AI and Machine Learning in Crisis and Disaster Management

AI applications for crisis and disaster contexts have matured rapidly, with substantial research activity across damage assessment, crisis communication, hazard forecasting, resource optimization, and search-and-rescue operations. Computer vision for building damage assessment represents one of the most developed application areas, anchored by benchmark datasets such as xBD, which provides over 850,000 building annotations across 45,362 km² of satellite imagery spanning six disaster types [Gupta et al. 2019]. Deep learning systems have progressed from convolutional neural networks through transformer-based systems achieving state-of-the-art performance, with operational deployments now supporting FEMA [Federal Emergency Management Agency 2025], the California National Guard, and humanitarian organizations [Bouchard et al. 2022; Kaur et al. 2023; Ronneberger et al. 2015]. Similarly, natural language processing for crisis informatics has evolved through over a decade of dataset development, from early lexicon-based approaches to domain-specific transformers that outperform general-purpose language models by 10–17% on crisis classification tasks [Alam et al. 2021a; Imran et al. 2014; Lamsal et al. 2024; Olteanu et al. 2014].

Studies that include professional EMs as research participants are relatively rare in the crisis informatics literature. Purohit et al. is a noteworthy exception, conducting a larger scale survey of U.S.- and Europe-based EMs about their social media use in work contexts [Purohit et al. 2025]. Studies that engage practitioners directly note organizational barriers to social media adoption and a pragmatic orientation toward “good enough” solutions given resource constraints [Purohit et al. 2025; Tapia and Moore 2014].

3 Methods

We conducted semi-structured, in-depth interviews with 32 U.S.-based EMs to understand their technology practices, AI experiences, and perspectives on future AI applications. Our study had three phases: (1) eligibility screening, (2) a pre-interview questionnaire collecting demographics and AI familiarity, and (3) in-depth virtual interviews (60–90 minutes). All materials received Institutional Review Board approval under exempt status. Extended methodological details, including research questions and instrumentation, are given in the Appendix D.

Participants. Participants were adults with two or more years of U.S. EM experience. We recruited participants using purposive sampling with snowball recruitment [Patton 2015]. To maximize diversity, we used maximum variation sampling [Patton 2015] across geographic regions, EM jurisdictional levels (local, state, federal), and employer types (public, private, nonprofit, academic). Our final sample (N=32) had 5–45 years of EM experience (M=20.7, Mdn=21.5) and was highly educated (50.0% master’s degrees, 28.1% bachelor’s, 12.5% doctoral). Full demographics appear in Table 1 and Table 2.

Interview Protocol. The protocol was developed iteratively from our research questions with EM practitioner-faculty coauthors to ensure domain-appropriate framing (Appendix D.4). We structured the flow of questions using a *funnel approach* [Morgan 1998], beginning with their EM background and disaster response experience and the details of their current job before introducing technology or AI topics. For substantive questions, we employed *scenario-based elicitation* drawing on the critical incident technique [Flanagan 1954], asking participants to walk through specific job scenarios before evaluating where technology might help.

Data Collection. The first author conducted all interviews via Zoom between March–April 2025, yielding approximately 41 hours of audio (procedures detailed in Appendix D.5). Recordings were transcribed using Otter.ai (institutional license), reviewed for accuracy, and de-identified with participant IDs reflecting state and sequence (e.g., FL3, MI1). The full interview protocol appears in Appendix E.

Data Analysis. We analyzed transcripts using reflexive thematic analysis [Braun and Clarke 2006, 2019]. The first author conducted open coding in ATLAS.ti, generating 847 initial codes capturing semantic content (e.g. “uses ChatGPT for message drafting”) and latent meaning (e.g. “frustration with legacy systems”). The second and third authors independently read a subset of the transcripts then met with the first author to discuss emerging

patterns. Through two additional rounds of iterative coding using constant comparison [Miles et al. 2014], we synthesized codes into candidate themes, created a codebook (Table 3), and iteratively updated the codebook after each round. This approach, guided by our codebook, yielded the components of Sections 4–5: contextual factors, AI perceptions, and 7 functional use case categories comprising 136 distinct AI and technology use cases.

4 Findings: EM Context and Perceptions of AI

By analyzing the 32 interview transcripts, we identified seven categories of AI Use Cases (UCs) corresponding to EMs' core job functions. Individual use cases support more granular job functions or actions that directly support those functions. While identifying these Technology Use Cases, we also captured the challenges EMs face in adopting technologies. Below, we present these findings in detail. First, we discuss the systemic challenges—independent of technology—that shape US EMs' work. Next, we highlight EMs' opinions on the role of AI in their profession.

4.1 Non-technical Systemic Challenges Significantly Impact EMs' Tech Adoption

U.S. EMs operate under unique constraints that fundamentally shape their technology requirements. Participants consistently reported working in “blue sky” conditions—non-activated periods that comprise approximately 95% of their time—yet current technologies focus disproportionately on active response. This misalignment between daily needs and the rigid design of available tools emerged as a key theme of participants' frustration. The field remains heavily text-driven, with standardized forms (e.g. FEMA ICS-201 through ICS-215) governing information flow across EM disaster phases.

Resource constraints emerged as the dominant theme across all 32 interviews:

- **Severe understaffing:** MA2, as the sole EM for an entire branch of their state's government, exemplified an extreme case: “I use AI all the time since I am a department of one [for my entire state].”
- **Multiple roles:** Participants averaged 3-4 different positions throughout their careers, with frequent “dual-hatting” during their current employment.
 MI1 captured a critical reality: “EM...is a side job for many people, and most of them don't have substantial [field] experience in EM.”
- **Knowledge silos:** Senior employees become single points of failure, with institutional knowledge and critical partner relationships concentrated in individuals rather than systems or processes
- **Budget limitations:** Limited funding for personnel and technology infrastructure

4.1.1 Current EM Technology: Universal Frustration with Legacy Systems. All participants with public sector experience reported negative experiences with WebEOC, the dominant EOC software used by state and local governments [Federal Emergency Management Agency 2019; Juvare, LLC 2025]. PA5 stated: “It requires too many clicks to do anything, and is not intuitive.” PA10 expanded on usability issues: “Limited utility and poor user interface, with too many categories and factors.” Beyond WebEOC, participants reported using a fragmented ecosystem of tools (Table 4).

4.1.2 Emerging AI Adoption: LLMs as a Gateway Technology. Despite limited professional exposure to AI, 23 of 32 participants (72%) had experimented with AI tools, exclusively Large Language Models (LLMs) and primarily ChatGPT. Current usage clustered around four specific applications:

- (1) **Public communication drafting** (18 participants): Creating messages at appropriate reading levels, with IL1 noting success in “drafting public messages at fifth-grade reading level as recommended by FEMA.”
- (2) **Exercise development** (12 participants): when designing training exercises and simulations for practicing readiness, FL3 described using ChatGPT to “create realistic exercise scenarios and injects that adapt based on player decisions.”

- (3) **Administrative writing** (11 participants): Email composition, report drafting, grant applications.
- (4) **Quick information retrieval** (8 participants): Processing unstructured queries like “list all zip codes for these counties.”

FL4’s assessment exemplifies the value of ChatGPT among participants who use the software: “ChatGPT cuts out a lot of work.”

4.2 Participants’ Perceptions of AI

Participants expressed overwhelmingly positive attitudes toward AI technology and its potential to support EMs in their work, with DC1 identifying AI as “transformative” and “the future of disaster management.” 31 of 32 participants (96.8%) expressed a net positive perception of AI’s current or future potential to support one or more EM job functions. Participants acknowledged AI’s potential as a tool to help alleviate their administrative burdens.

EM-focused AI Systems should preserve human autonomy. Participants unanimously agreed that EM AI systems should preserve human decision-making authority. Feng et al. propose a framework outlining 5 levels of human versus AI agent autonomy in AI systems. We used this framework to benchmark participants’ responses about their preferred user’s autonomy for EM AI systems. According to that framework, participants’ preferences align with the following two configurations: *User as Operator (L1): AI provides information or analysis, but humans make all decisions and take all actions*; and *User as Collaborator (L2): AI suggests options or drafts outputs, but humans review, modify, and approve before implementation*. No participants expressed interest in AI systems that would make autonomous decisions without human oversight, particularly for life-safety applications. FL2 explained that, regardless of whether or not they use AI to make a decision, EMs are still accountable for the outcome, subject to other government and stakeholder oversight: “...at the end of the day, I’m the one who has to stand in front of the county commissioners and explain why we did what we did.”

Pragmatic Tolerance for Imperfection. Participants demonstrated willingness to work with imperfect AI systems—a pragmatism developed through years of using inadequate technical tools. MI1 articulated this perspective: “Look, nothing we use now is perfect. If AI can get me 80% of the way there and I can fix the rest, that’s still a huge time savings.” This tolerance extended to accepting occasional errors, provided the AI’s reasoning was transparent and outputs could be verified. Participants consistently emphasized the need to understand AI reasoning because they remain accountable for decisions regardless of AI involvement. PA1 stated: “EMs need to know all the minute details. If something goes wrong and I have to explain it, ‘the AI told me to’ is not going to fly.” This suggests that explainability and auditability may be more important design priorities than marginal accuracy improvements for EM applications.

AI as a Potential Tool for (In)equity. Several participants articulated how AI might address—or inadvertently exacerbate—existing inequities in EM. AZ1 surfaced existing bias in disaster response resource allocation, noting that decisions are sometimes influenced by “people’s last names,” housing type, and neighborhood characteristics. AZ1 envisioned AI as a tool that could “help eliminate bias in some way to be more equitable in terms of [resource] prioritization,” referencing the US CDC’s Social Vulnerability Index (SVI) as a common data source used by EMs to identify under-resourced communities.

LA1 identified bias in existing damage assessment processes: “damage gets underestimated [in less affluent neighborhoods], because there’s sort of the assumption that they weren’t that nice to begin with.” LA1 also noted a concerning pattern in technology transitions: when EMs adopt new communication tools (e.g., text alerts), older channels such as mail and in-person notifications are often deprecated, “leaving behind the population that uses those channels.”

NC1 described incorporating equity considerations into tactical response planning by consulting “a social vulnerability index [SVI] map, a climate justice map” when making resource allocation decisions. These perspectives suggest that participants view AI not merely as an efficiency tool but as a potential mechanism for either reinforcing or mitigating systemic inequities in disaster response.

Summary. In an environment of chronic resource constraints (most profoundly in public sector and local contexts), US EMs demonstrate frustration with existing technology and pragmatic enthusiasm for domain-specific AI tools that retain their ultimate decision-making autonomy. In Section 5, we describe seven functional technology use case categories in EM that reflect varying levels of technological maturity and AI readiness. For each use case, we examine the feasibility of participant-envisioned AI applications and identify challenges for near-term development. Section 6 will then elaborate on cross-cutting design implications for developing EM AI tools and the opportunities we believe are the most promising for future research.

5 Analysis of EM AI Use Cases

Our analysis identified 136 distinct technology use cases across the seven categories of EM operational capabilities. We present these ordered by the number of participants who mentioned use cases in each category, as this reflects both current pain points and perceived opportunities for technological intervention. For each use case, we describe (1) the current state of technology and practices, (2) key challenges participants identified, (3) current AI use if any, and (4) envisioned AI applications. The feasibility assessments in this section draw on author expertise in EM practice, responsible AI research, and robotics, complemented by technical literature review (expanded in Section 2). These assessments represent our analysis of alignment between participant-envisioned applications and current technological capabilities—they are not empirical findings from participant data. We distinguish between participant-reported experiences (e.g., “Current State,” “Current AI Use”) and our analytical contributions (e.g., “Feasibility Assessment,” “Technical Foundations,” and “Key Challenges”) to maintain transparency about the methodological basis of each claim. To make this distinction visible at a glance, labels for participant-reported content are set in **bold** and labels for our analytical contributions are set in **colored bold** throughout this section.

5.1 UC1: Public and Stakeholder Messaging (24 participants, 75%)

Public and stakeholder messaging was the most frequently-mentioned UC, reflecting both its high operational burden and practitioner’s experimentation with general-purpose AI for this task. This is likely related to the popularity and accessibility of LLMs and how well their capabilities align with messaging tasks—though important limitations remain.

Current State. Emergency (warning and alerts) and preparedness messaging is a crucial part of EMs’ responsibility to engage the whole community [Yeung and Clark-Ginsberg 2024] in EM. This involves managing communications across multiple platforms—social media, mass notification systems [U.S. Department of Homeland Security 2025], email, press releases, and direct outreach—intended for a variety of audiences. **IL1**’s experience with routine siren testing illustrated recurring communication burdens: “So Tuesday, we test our outdoor warning sirens first Tuesday of every month at 10am...Inevitably, I always get a call...from somebody in the public about the outdoor warning siren.” Such predictable, repetitive communication tasks exemplify opportunities for automation.

Current AI Use. Multiple participants reported using ChatGPT for message drafting. **IL1** specifically used it to draft short-form posts to “inform [the] public about function of outdoor warning sirens,” while **MA2** used it for “translating technical weather bulletins into plain language warnings.”

Opportunities for AI Development:

- *Audience-specific message customization:* Automatically adjusting message complexity, readability, and channel of a message based on the expected audience’s demographics

- *Multi-language translation*: Real-time translation for diverse communities
- *Message compliance checking*: Evaluate whether a particular message draft is consistent with applicable mass notification (and related) best practices
- *Misinformation monitoring*: AI systems that detect and flag false information requiring official correction

For near-term implementation, template-based message generation systems that maintain organizational voice while adapting complexity for different audiences, hazards, and channels—combined with human review workflows and alignment checks against mass notification best practices—represent the most tractable path forward.

Feasibility Assessment: High for core tasks, low for misinformation detection and information verification.

Technical Foundations. Message drafting, editing, and reading-level adaptation represent genuinely tractable problems. LLMs excel at reformulating or summarizing detailed or technical content for lay audiences, and participants already report success using ChatGPT for this purpose. The infrastructure requirements are relatively minimal—a web browser and internet connection—making adoption feasible even for severely resource-constrained agencies. Using publicly-available AI tools (like ChatGPT) does not present a significant risk of data leakage, as the message content for public audiences by nature should not contain sensitive or otherwise classified information. Multi-language translation similarly leverages mature NLP capabilities, with commercial translation APIs achieving professional-grade quality for common language pairs like English-Spanish.

Key Challenges. Participant-envisioned capabilities around misinformation monitoring warrant significant caution. Automated detection of false information during crises presents well-documented challenges: misinformation often spreads faster than verification is possible, classification accuracy drops substantially for novel claims, and false positives risk undermining legitimate community-sourced intelligence [Arif et al. 2016; Starbird et al. 2014]. The verification challenge is compounded in emergency contexts where ground truth itself may be uncertain or evolves at a much faster pace than data pipelines and/or downstream AI models are updated.

5.2 UC2: Situational Awareness (15 participants, 47%)

Situational awareness use cases focused on synthesizing multiple information streams into actionable intelligence during both steady-state monitoring and active response.

Current State. EMs currently rely on manual monitoring of multiple channels simultaneously. Verification and integration of crowd-sourced reports is manual and time-intensive.

Current AI Use. None observed. Participants did not report using AI tools for situational awareness functions.

Opportunities for AI Development. Participants identified three primary opportunities:

- *Social media intelligence*: Extracting damage reports and public sentiment from platforms, with geographic clustering to identify “hot spots”
- *Multi-stream synthesis*: WW1’s request for “unifying data streams” reflected widespread desire for AI dashboards combining weather, infrastructure, and social data
- *Anomaly detection*: Automatic flagging of unusual patterns requiring human attention

PA4 described an innovative crowdsourcing approach: “[I would like a] virtual emergency operations center where we could get vetted public to say, ‘Hey, this is happening.’...[then] this goes right to our virtual information gathering.”

Feasibility Assessment: Moderate, with significant infrastructure barriers.

Technical Foundations. Crisis informatics research has developed sophisticated approaches to social media filtering and classification [Alam et al. 2021a; Imran et al. 2014; Lamsal et al. 2024], anomaly detection algorithms can identify unusual patterns in sensor data, and dashboard technologies can visualize diverse data sources. The CrisisTransformers family of models demonstrates that domain-specific NLP can substantially outperform general-purpose systems for crisis-related text [Lamsal et al. 2024].

Key Challenges. Translating research capabilities into useful operational tools faces substantial barriers. First, *data access*: blanket integration of 911 dispatch feeds, weather services, social media streams, and field reports requires API access, interoperability protocols, and institutional agreements that exceed most local agencies' technical capacity. Second, the “*degraded conditions*” problem identified by participants means that sophisticated cloud-based tools may be unavailable precisely when most needed—during active response efforts when infrastructure fails. Third, *verification remains challenging*; participants emphasized wanting AI to process social media while simultaneously avoiding false information, a tension without easy resolution.

5.3 UC3: Planning (12 participants, 38%)

Planning emerged as a promising area for AI application due to abundant existing data and clear evaluation criteria. The EM Accreditation Program (EMAP) provides standardized benchmarks [Emergency Management Accreditation Program 2022], while thousands of publicly available plans offer potential training data.

Current State. Planning involves creating and maintaining emergency operations plans, continuity of operations plans, and hazard mitigation plans that must comply with EMAP standards. FL4 provided an example of a typical problem scope for an EM whose jurisdiction is a large school district: “We had emergency plans that all of our schools were responsible for creating...our K-8, our tech colleges, our administrative sites, they all had to have an emergency plan,” and it was the EM’s responsibility to verify that those plans are up-to-date, in compliance with relevant regulations, and do not conflict or interfere with other facilities’ plans. To achieve this, EMs perform manual review and cross-referencing of these plans, which consumes significant time.

Current AI Use. None observed. Participants did not report using AI tools for planning functions.

Opportunities for AI Development. Participants described four primary enhancements:

- *Automated plan review*: Evaluating plans against established standards and identifying gaps
- *Dynamic update recommendation*: Identify plan components that may be impacted by input demographic or infrastructure changes to jurisdiction
- *Dependency detection (crosswalk)*: Identifying resource conflicts, such as multiple facilities using the same supplier
- *After Action Review synthesis*: Analyzing reviews from similar jurisdictions to suggest improvements

Feasibility Assessment: High.

Technical Foundations. Several factors make planning unusually well-suited to current AI capabilities. First, *data availability*: thousands of emergency operations plans, continuity of operations plans, and hazard mitigation plans are publicly accessible and could be used for training corpora. Second, the Emergency Management Accreditation Program (EMAP) provides standardized criteria against which plans can be evaluated, enabling objective assessment of plan completeness and compliance. Third, planning occurs during “blue sky” conditions when connectivity is reliable and time pressure is reduced, avoiding the urgency constraints that complicate response-phase applications. Participants’ desired applications align well with document analysis and retrieval-augmented generation capabilities. Additionally, the text-dominant nature of planning documents complements LLM strengths.

Key Challenges. The primary challenge is temporal obsolescence, as identified by CA1 : “Plans take a long time to develop. By the time they are finalized, many things about the region the plan was made for change substantially.” Static plans become obsolete quickly, yet updating requires repeating the entire planning process. To ensure plan viability, EMs currently conduct manual review and cross-referencing of these plans—a practice that consumes significant time. Additionally, comprehensive dependency detection—identifying resource conflicts such as multiple facilities planning to use the same supplier in a disaster—requires integration across documents that may not be digitized or standardized.

5.4 UC4: Incident Management (11 participants, 34%)

Incident management encompasses EOC activation decisions, resource coordination, and operational tempo maintenance.

Current State. Incident management operates through the Incident Command System (ICS), with standardized resource typing enabling coordination across jurisdictions. FL2 explained the typing system: “Rather than saying ‘I need an ambulance,’ you say ‘I need a Type III EMS team,’ which includes one ambulance, six people to run it 24 hours, supplies, fuel, and maintenance for seven days.” Documentation occurs through Incident Action Plans (IAPs) that capture the Incident Commander’s intent and operational details.

Current AI Use. None reported.

Opportunities for AI Development. (1) *Resource matching*: AI systems that facilitate resource matching and availability tracking across jurisdictions using standardized typing. (2) *Incident Action Plan generation*: FL2 identified a specific opportunity: “If we had a way where we could provide the Incident Commander’s intent, we could supply bulleted items, and then the AI flushes that out and creates an actual document...that’s a draft that can be reviewed.”

Feasibility Assessment: Mixed—high for documentation, low for activation decisions.

Technical Foundations. Documentation and resource coordination applications can effectively leverage existing LLM capabilities. Incident Action Plan generation—where an Incident Commander provides intent and key points, and a domain-specific model can expand these into formatted documents—uses LLM capabilities while maintaining human authority over operational decisions. Similarly, standardized resource typing creates structured data amenable to AI-driven analysis.

Key Challenges. Multiple participants expressed interest in AI assistance for deciding when to activate an EOC, but this represents a rabbit hole that computing researchers should avoid in the near future. Deciding when to activate an EOC is not low hanging fruit; there are 1400 different counties in the United States, and every single one of them would probably have different activation criteria. Activation decisions involve liability considerations, elected official preferences, political pressures, resource availability, and community expectations that cannot be captured in algorithmic rules. Attempting to automate or even recommend activation decisions risks creating tools that are either too rigid to accommodate local variation or too vague to provide actionable guidance.

5.5 UC5: Damage Assessment (10 participants, 31%)

Damage assessment represents the most technologically mature use case category, with existing computer vision applications and benchmark datasets enabling relatively straightforward translation of research capabilities to operational tools. However, participants identified significant gaps in processing and analysis.

Current State. Drone technology with damage assessment, such as the “Skydio drone with Fusus for damage assessment” used by FL6. However, current workflows remain largely manual. MI1 described the limitation: “It’s us manually just patrolling the area with drones, looking with our eyes for anything.”

Current AI Use. Only one explicit AI-powered usage (FL6). 10 participants mentioned using basic recording features of commercial drones to collect data.

Opportunities for AI Development. Participants described four primary enhancements:

- *Automated damage detection*: Identifying structural damage, debris fields, and affected infrastructure
- *Change detection*: Comparing pre- and post-event imagery to quantify impact
- *Generating documentation*: Automatically generating standardized Preliminary Damage Assessments with photographic evidence
- *Search target identification*: Flagging potential victims or specific objects (vehicles, debris) in UAV footage

Feasibility Assessment: High for detection, moderate for integration.

Technical Foundations. The primary bottleneck participants identified—manual review of extensive drone footage—is precisely a task at which current computer vision methods can perform relatively well.

Key Challenges. The primary bottleneck is the volume of data required for analysis. Another challenge lies in integration rather than core AI capabilities. For the highest-value and long-term sustainability of data quality, drone footage should be geolocated, AI system outputs connected to GIS systems for visualization, and results should feed into existing damage assessment workflows. Though these are primarily engineering challenges rather than fundamental research problems, they require attention to workflow integration and interoperability.

6 Discussion

Our findings reveal EM as a domain with exceptional opportunity for impactful AI research and development, grounded in high stakeholder need due to long-standing resource constraints and general openness to more feature-rich but imperfect tools. The use cases identified by participants represent a research agenda for the computing community. This section examines the implications of these findings for AI research and design, identifies opportunities for near-term impact, and addresses constraints and complications that researchers must navigate.

6.1 Implications for EM AI Research and Design

The EM context challenges several assumptions embedded in conventional approaches to AI system design. We identify five design principles grounded in our findings.

6.1.1 Design for Role Fluidity. Traditional user-centered design assumes relatively stable user roles and tasks. EM, especially in local and “department-of-1” settings, fundamentally violates this assumption. The same individual might serve as planner, public information officer, and operations chief within a single activation. Participants described averaging 3–4 different positions throughout their careers, with many public sector employees “dual-hatting” during their employment. Systems must therefore support rapid role-switching without extensive reconfiguration. Interface designs that require users to customize detailed profiles or settings before use will fail when an EM assumes a new role during an escalating incident. Instead, systems should offer role-appropriate defaults that can be quickly adjusted, with core functionality accessible regardless of the user’s current operational position.

6.1.2 Prioritize Cognitive Load Reduction. High cognitive demands emerged as the core challenge across all use cases. **MI1** recalled an instance of tornado response experience—simultaneously monitoring police radio, 911 dispatch, National Weather Service updates, social media reports, and field personnel communications—exemplifies a universal struggle: participants consistently described “drowning in data” without synthesis tools. Rather than adding more data streams, successful EM technologies should reduce cognitive load through intelligent filtering, prioritization, and summarization. Systems that incorporate more information sources without helping users process existing streams risk exacerbating rather than alleviating the fundamental problem.

6.1.3 Design for Degraded Conditions. EM response and recovery operations often occur when infrastructure fails. Power outages, cellular network interruptions, and loss of Internet connectivity frequently coincide with disasters and incidents that warrant emergency response. A key requirement for effective systems is that they must function with intermittent connectivity, partial data, and stressed users. Researchers should explicitly design systems to accommodate degraded conditions—reduced bandwidth, intermittent connectivity, incomplete data feeds—rigorously test their performance in comparable settings.

6.1.4 Rigorously Define and Support “Good Enough” AI Solutions. Participants demonstrated willingness to work with imperfect AI systems—a pragmatism developed through years of using inadequate tools. Tapia and Moore’s

concept of “good enough” solutions for EM resonates strongly with our findings. This mindset tolerates systems that would be considered insufficiently accurate in other domains, provided there is appropriate human oversight of outputs. Systems should also actively engage human users in critically reviewing and interrogating AI outputs rather than encourage passive acceptance.

6.1.5 Incorporate Accountability Structures and Mechanisms for Equitable Outcomes into Design. EMs operate within accountability structures—to elected officials, to oversight bodies, to affected communities—that shape how AI systems can be responsibly deployed. Participants’ emphasis on understanding AI reasoning reflects not abstract curiosity but practical necessity: they must be able to explain and defend their decisions. AI systems that undermine this accountability—by providing recommendations without transparent reasoning, by encouraging passive acceptance of outputs, or by diffusing responsibility for decisions—risk both operational failures and erosion of democratic oversight. The FAccT community’s work on algorithmic accountability [Alkhatib and Bernstein 2019; Saxena et al. 2021] provides frameworks directly applicable to EM contexts.

Beyond procedural accountability, AI systems in EM must consider how AI-informed decisions may impact equitable real-world outcomes. As our findings reveal (Section 4.2), participants identified existing biases in resource allocation and damage assessment that AI could either perpetuate or help mitigate. AZ1’s observation that decisions are sometimes influenced by “people’s last names” and neighborhood characteristics, and LA1’s concern that “damage gets underestimated [in less affluent neighborhoods],” highlight that EM decisions have profound equity implications: which communities receive warnings first, how resources are allocated across neighborhoods, whose damage reports are prioritized for assessment. Drawing on Alkhatib and Bernstein’s concept of “street-level algorithms” and Saxena et al.’s framework for high-stakes algorithmic decision-making in the public sector, we argue that responsible EM AI development requires explicit attention to:

- *Differential performance across communities:* Training data biases—the geographic limitations of damage assessment datasets, the English-centricity of crisis NLP corpora—risk creating systems that perform poorly for already-marginalized communities. Systems should be evaluated for performance disparities across demographic and geographic subgroups.
- *Feedback loops and resource allocation:* AI systems that inform resource allocation decisions may create feedback loops where underserved communities receive less attention, generating less training data, leading to poorer model performance for those communities.
- *Human oversight as equity safeguard:* Research on the Allegheny Family Screening Tool demonstrates that human-AI collaboration can reduce rather than amplify disparities when systems are designed appropriately [Cheng et al. 2022]. Similarly, it is essential to leverage human judgment to review outputs and help correct for algorithmic limitations.
- *Deprecation effects:* When new technologies are adopted, older channels are often deprecated, potentially leaving behind populations who relied on them (LA1). AI system deployment should consider who may be excluded by technological transitions.

6.2 Near-term Research and Development Opportunities

Based on our feasibility analysis and participant priorities, we identify three areas that offer immediate opportunities for research and development. These represent our assessment of where current AI capabilities most closely align with practitioner needs and where meaningful progress is achievable in the near term.

Public Messaging Drafting and Revision. Message drafting represents perhaps the most immediately tractable application area. Participants already report using general-purpose LLMs for drafting messages; compared to other UCs, the infrastructure requirements are relatively light (processing short-form text is less compute- and data-intensive). Research opportunities include developing EM-specific prompt libraries, evaluating message

effectiveness across different audience segments, establishing best practices for human review workflows, and—critically—examining equity implications of automated translation and audience targeting (e.g., which languages are prioritized, how reading-level adaptation may inadvertently exclude or patronize certain communities).

Exercise Generation Platforms. The “safe failure” environment of exercises offers ideal conditions for AI experimentation. Systems generating adaptive scenarios with performance analytics could improve preparedness while simultaneously building practitioner trust in AI capabilities. As FL2 observed, exercises exist “because that’s the perfect time to make mistakes”—this philosophy extends naturally to AI integration. Exercises provide opportunities to test AI capabilities, calibrate user trust, identify failure modes, and refine systems before operational deployment. Research opportunities include developing adaptive scenario engines, automated performance capture tools, and after-action report synthesis systems.

Plan Analysis Tools. The combination of abundant data (thousands of publicly available emergency plans), clear evaluation standards (EMAP criteria), and the text-native nature of planning documents makes this domain well-suited for current AI capabilities. Automated plan review, gap analysis, and update recommendations could significantly reduce the planning burden that consumes substantial portions of EMs’ “blue sky” days. The temporal challenge identified by CA1—that plans become obsolete before finalization—could be partially addressed by AI systems that monitor for changed conditions and identify sections that may be impacted by those changes.

7 Limitations and Future Work

Our study focused on U.S. EMs, who operate in a landscape of overlapping local, regional, and federal regulations specific to the U.S. and their local EM jurisdictions—limiting generalizability to other contexts. International EM operates under different frameworks (e.g., the UN cluster approach) that might yield different requirements.

Additionally, our interview questions (Appendix E) focused on EMs’ imagined and anticipated uses of AI rather than their actual experiences deploying existing AI systems. We attribute this to several causes. First, there is limited availability of EM-specific AI tools—and an even more limited set of EM departments able to afford such sophisticated production software [Argonne National Laboratory and National Preparedness Analytics Center 2025; Deloitte Center for Government Insights and National Emergency Management Association 2025]. Second, in the recruitment phase of our interview study (Appendix D.2), we deliberately did not advertise or seek out EMs with AI experience.

Furthermore, this study focused on participants’ challenges and needs at work; as a natural consequence, participants primarily discussed AI as a tool that might support or expedite specific EM job functions. This functional tool-oriented framing of AI is appropriate for eliciting design recommendations as it is grounded in practitioners’ concrete operational needs. However, it does not fully capture AI as a sociomaterial phenomenon that can reshape organizational practices, professional identities, and institutional relationships. Future work should complement our task-oriented findings with analyses that examine the broader sociotechnical effects that AI adoption may catalyze in EM.

Our interview protocol (Appendix E) does not explicitly probe the equity dimensions of AI in EM. Though participants organically raised equity-related concerns (as reported in Section 4.2), we did not include structured questions on pertinent topics, such as how AI systems might differentially affect historically marginalized communities or how practitioners reason about fairness when envisioning AI tools. As a result, our findings may underrepresent the equity considerations that should inform responsible AI development for EM.

Finally, we deliberately do not provide a prescriptive mapping of the identified UCs to specific AI system types or taxonomies. This choice reflects two considerations: first, our UCs were elicited from EMs who are not AI experts and who, therefore, did not describe their needs with the technical granularity required for precise AI system specification. Second, the primary scope of this work is to surface the UCs that practitioners themselves identify as important, establishing a foundation for future work to align AI development with real-world operational

needs. Translating these UCs into specific AI system requirements represents a distinct research contribution that should involve both AI researchers and EM practitioners in participatory design processes.

Future Work. Dedicated study of ethical dimensions of AI in EM represents a fruitful and timely direction for future work. This could include examining how AI tools might affect resource allocation across communities and whether AI-generated recommendations reproduce or mitigate existing disparities in disaster outcomes.

Additionally, future work should investigate how AI tools can be grounded in the larger social context of EM, examining not only what AI tools EMs need, but how AI reshapes organizational practices, inter-agency coordination, and equity outcomes. Prior work on street-level algorithms [Alkhatib and Bernstein 2019] and high-stakes algorithmic decision-making in the public sector [Saxena et al. 2021] provides methodological models for this research direction. Such studies should examine how AI systems interact with the resource constraints, accountability structures, and community relationships that characterize EM work.

As AI-driven technology for EMs matures, we envision two important threads for future work studying EM AI adoption. The first focuses on the impact of EM-specific AI tools that examine the driving factors behind adoption outcomes. Such work would provide insights into how to design better tools. Longitudinal studies could track whether specific AI solutions play a role in improving EMs' user experience with domain-specific technology or other widespread challenges in the field such as those in Section 4.1. Second, the organizational, institutional, and political implications of AI adoption in EM represent a complementary research direction motivated by our findings and participants' expressed needs. Our interviews surfaced enthusiasm for AI tools but also revealed complex institutional dynamics—relationships with elected officials, inter-agency coordination challenges, and community trust considerations—that will shape how AI is actually adopted and used. Understanding these dynamics requires different research methods, such as organizational ethnography, policy analysis, downstream impact assessments, and longitudinal studies of AI implementation in EM agencies. These would complement our practitioner-centered findings with the institutional perspective necessary for responsible AI deployment in EM.

8 Conclusion

Our investigation of 32 EMs' technology practices and AI perspectives reveals a profession uniquely positioned to benefit from AI advancement. EMs demonstrated pragmatic enthusiasm for AI adoption [Yu et al. 2025] driven by chronic understaffing and frustration with existing tools. Participants viewed AI not as a threat to their jobs but as a potential force multiplier that might help them accomplish work that currently goes undone. At the same time, they emphasized the importance of maintaining human decision-making authority, understanding AI reasoning, and designing for the degraded conditions under which incident response actually occurs. Through analysis of 136 technology use cases, we identified seven categories of operational needs, with public messaging, situational awareness, and planning emerging as prime candidates for near-term AI application.

The EM context challenges conventional HCI assumptions about user roles, task stability, consistent operating conditions, and the primacy of accuracy over explainability. Successful EM technologies must support radical role flexibility, reduce rather than increase cognitive load, and function under degraded conditions. These findings carry implications for computing researchers and practitioners interested in developing AI for high-stakes, resource-constrained domains. By centering practitioner perspectives before AI systems are built rather than after they fail, this study offers guidance for responsible AI development in a domain where the stakes—and the opportunities for meaningful impact—are exceptionally high.

Most critically, our findings reframe AI's potential role in EM from an amorphous threat to an essential force multiplier and collaborator. In a field where "department of one" scenarios are commonplace and lives depend on rapid decision-making, AI represents not job displacement but a critical tool—for both EMs and the communities they protect. As climate change increases disaster frequency and intensity while resources remain constrained, developing effective AI tools for EM becomes not just an interesting research problem but a societal imperative.

Generative AI Usage Statement

We used Generative AI (Claude Opus) to generate $\text{\LaTeX}$ tables of participant demographics and methods artifacts from pre-existing spreadsheets and Overleaf’s AI-powered grammar engineer to identify minor typos and typesetting errors in the paper.

Positionality Statement

Our team combines expertise across machine learning, responsible AI, human-computer interaction, disaster robotics, and emergency management. Our EM coauthors (McDaniel and Merrick) are practitioner-faculty who have collectively participated in hundreds of crisis situations and held command staff positions during major disasters. They shaped our interview protocol, domain framing, use case analysis, and feasibility assessments. Foubert drew on her background in responsible AI research and ML engineering to lead the end-to-end study design, investigation, and analysis. Prior to and during the study, she participated in informal field observations with EM practitioners, including ride-alongs, EOC visits, and tabletop exercises—and engaged with online EM professional communities to develop contextual familiarity with the domain. Park contributed HCI and design expertise that shaped our analysis of user-facing design implications, including the design principles in Section 6. Murphy and Manzini bring extensive experience in disaster robotics and field deployments with practitioners; they contributed to our use case analysis and feasibility assessments. Singh, Pardeshi, and Heidari provided expertise in AI, statistical machine learning, and AI fairness, respectively, grounding our feasibility assessments and equity analysis.

This study focuses on U.S.-based emergency managers operating within U.S. doctrine (NIMS, ICS, FEMA) and the conventions we discuss reflect that scope. International EM operates under different frameworks, and our findings should be interpreted accordingly. Our use case assessments draw on the team’s interdisciplinary expertise rather than an exhaustive cataloging of every use case surfaced in interviews; we prioritized use cases where we could offer informed analysis of the alignment between practitioner needs and current AI capabilities.

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A Participant Information

Table 1. Participant Data. Each row represents one of 32 interview participants, showing their EM role type, anonymized ID, jurisdiction scope, location in the U.S., years of EM experience, self-reported AI familiarity, and frequency of AI use in general and in EM work contexts. Role types reflect participants' current professional functions: **Research** includes university faculty and full-time researchers; **Leadership** includes county and state EM directors, EOC managers and other EMs in leadership roles; **EM Partner** includes fire chiefs, public safety directors, and nonprofit disaster services leaders who coordinate closely with EM; **Practitioner** includes EM contractors, business continuity managers, EM specialists in healthcare and corporate settings, and subject matter experts. Combined categories (e.g., **Research & Leadership**) indicate participants whose roles span both functions.

Type of EM	ID	Jurisdiction Scope	Location	Years	AI Fam.	AI Usage Frequency	
						General	EM work
Research	LA1	N/A (Researcher)	LA	20	A little	Rarely	Rarely
	WW1	National	VA	21	A lot	Often	Rarely
	AZ1	N/A (Researcher)	AZ	5	A little	Often	Often
Research & Leadership	FL2	State, Local (Other)	FL	22	A lot	Rarely	Rarely
	MI1	Local (County)	MI	16	A lot	Often	Often
Leadership	PA4	Local (County)	PA	33	A lot	Very freq	Very freq
	DC1	State	DC	12	A lot	Often	Never
	NC1	Local (Municipal)	NC	12	A little	Rarely	Rarely
	PA2	Local (County)	PA	26	A lot	Rarely	Rarely
	FL3	Local (Other)	FL	15	A lot	Very freq	Very freq
	IL1	Local (County)	IL	22	A little	Often	Often
	PA8	Local (Other)	PA	25+	A little	Often	Rarely
	PA1	Local (County)	PA	13	A lot	Often	Rarely
	TX1	Contractor	TX	45	A lot	Often	Often
	MA2	State	MA	19	A lot	Often	Often
	PA7	Local (Other)	WV, PA	26	A little	Rarely	Never
EM Partner & Leadership	PA10	Local (Municipal)	PA	38	A little	Rarely	Rarely
	PA3	Local (County)	PA	6	A little	Rarely	Rarely
	NW4	National	IL	25	A lot	Often	Often
EM Partner	PA6	Local (County)	PA	26	A lot	Often	Often
	FL6	Local (Municipal)	FL	35	A little	Often	Rarely
Practitioner	PA5	Local (Other)	PA	12	A lot	Very freq	Very freq
	WW2	Worldwide	ID	25	A little	Rarely	Never
	FL4	Local (Other)	FL	18	A little	Rarely	Rarely
	NY1	Local (Municipal)	NY	17	A lot	Often	Rarely
	NW2	Worldwide	ME	25	A little	Often	Often
	CA1	Contractor	CA	30+	A little	Never	Never
	NW3	Contractor	FL	12	A lot	Never	Never
	NW5	Contractor	NY	25	A lot	Very freq	Often
	DC2	Retired	PA	23	A lot	Very freq	Often
Practitioner & Research	FL1	State, Local (Other)	FL	9	A lot	Very freq	Never
	FL8	State, Local (Other)	FL	5	A lot	Often	Rarely

A.1 Participant Information

Table 2. Participant Demographics (N = 32). Aggregate demographic characteristics of the interview sample, including professional experience, age, gender, race/ethnicity, education, veteran status, and political orientation. Response rates for optional items ranged from 69% to 100%.

Characteristic	Value
<i>Professional Experience</i>	
Years of EM experience	Range: 5–45; M = 20.7, Mdn = 21.5
<i>Age Range</i>	
25–34	5 (15.6%)
35–44	7 (21.9%)
45–54	11 (34.4%)
55–64	5 (15.6%)
65+	4 (12.5%)
<i>Gender (n = 31)</i>	
Male	28 (90.3%)
Female	3 (9.7%)
<i>Race/Ethnicity (n = 30)</i>	
White	24 (80.0%)
Person of color	6 (20.0%)
<i>Education (n = 32)</i>	
Master’s degree	16 (50.0%)
Bachelor’s degree	9 (28.1%)
Doctoral degree	4 (12.5%)
Other (Associate, Professional, Some college)	3 (9.4%)
<i>Veteran Status (n = 22)</i>	
Veteran or current military/reserves	4 (18.2%)
<i>Political Orientation (7-pt. scale; n = 22)</i>	
Social issues	M = 4.0, Mdn = 4.0, SD = 1.4
Economic/fiscal issues	M = 4.5, Mdn = 5.0, SD = 1.5

B Additional Use Cases

B.1 UC6: Exercises (8 participants, 25%)

Exercise design and execution emerged as an ideal testing ground for AI applications, offering a “non-threatening” environment for experimentation without life-safety implications.

Current State. Exercises are an established component of emergency management practice, providing opportunities for training and plan validation. FL2’s insight was compelling: “...one of the reasons that Emergency Managers do a lot of exercises...is because that’s the perfect time to make mistakes. We want to walk ourselves into as realistic a scenario as we possibly can.”

Current AI Use. This category showed the strongest current AI adoption beyond public messaging. FL3 reported “creating community to expand AI utilization” for exercise development, while NW5 used AI to generate “realistic exercises with dynamic injects.”

Opportunities for AI Development. Participants described four primary enhancements:

- *Adaptive scenarios*: Exercise narratives that respond to participant decisions in real-time
- *Performance analytics*: Automated capture and analysis of decision points and outcomes
- *After-action synthesis*: Combining reports from multiple exercises to identify patterns
- *Decision branch exploration*: AI-generated alternative scenarios exploring “what-if” situations

Feasibility Assessment: High, with strategic value beyond exercises themselves.

Technical Foundations. Current participant applications demonstrate LLM utility for narrative generation tasks. More sophisticated applications could include adaptive scenarios that respond to participant decisions in real-time, automated performance analytics capturing decision points and outcomes, after-action synthesis combining reports from multiple exercises to identify patterns, and decision branch exploration through “what-if” scenario generation. These capabilities leverage LLM strengths while contributing to the training and preparedness functions that occupy a significant portion of EMs’ “blue sky” working hours.

Key Challenges. However, exercises must maintain sufficient realism and diversity to provide training value—AI-generated scenarios that are implausible or inconsistent may undermine learning objectives. In addition, automated performance analytics (such as for an After-Action Report) require careful design to capture meaningful metrics without creating adverse incentives or oversimplifying complex decision-making [Jacobs and Wallach 2021].

B.2 UC7: Search and Rescue (6 participants, 19%)

Search and rescue (SAR) UCs presented clear technological needs with direct life-safety implications.

Current State. Most SAR coordination involves basic tools—radio communications, paper maps, and manual search grids. Technology adoption in this area lags behind other use cases: NY1 described “coordinating multiple search teams across jurisdictions with incompatible radio systems.” Interoperability failures and fragmented communication infrastructure complicate multi-agency operations precisely when coordination is most critical.

Current AI Use. None observed. Participants did not report using AI for SAR functions.

Opportunities for AI Development. Participants identified three primary opportunities:

- *Search optimization*: AI-driven search patterns based on terrain, weather, and victim profiles
- *Social media scanning*: Identifying potential victim locations from posts and images
- *Multi-agency coordination*: Automated deconfliction of search areas and resource allocation

Feasibility Assessment: Moderate—technology exists, integration is challenging.

Technical Foundations. AI-enhanced search and rescue are technically well-established. Deep learning approaches to aerial imagery analysis achieve highly accurate real-time performance on embedded hardware suitable for Unmanned Aerial Vehicle (UAV) deployment [Božić-Štulić et al. 2019; Dong et al. 2021]. Reinforcement learning methods can optimize search patterns based on terrain, weather, and victim profiles [Ewers et al. 2025; Queralta et al. 2020]. Social media analysis could potentially surface victim location information from posts and images.

Key Challenges. Meaningful improvements to SAR technology face coordination and resource challenges that software alone cannot resolve. Currently, operations employ tools with little to no integration—radio communications, standard street/satellite maps, and manual search grids. This highlights that communication infrastructure—not AI capability—often represents the binding constraint. AI-optimized search patterns provide limited value if teams cannot receive or act on recommendations due to interoperability failures. Multi-agency coordination requires not just technical solutions but institutional agreements and standardized protocols that exceed the scope of AI development alone.

C Additional Related Work

C.1 Crisis Informatics

Crisis informatics has developed over two decades as a distinct research area within HCI and CSCW, producing substantial knowledge about information flows during disasters. The field's foundational orientation, however, has centered “citizen-side” perspectives—examining how affected populations and volunteers use information and communication technologies rather than how professional EMs accomplish their work [Palen and Anderson 2016; Palen and Liu 2007]. Seminal studies documented how community members self-organized via social media following the Virginia Tech shooting [Palen et al. 2009], characterized public “backchannel” communications during California wildfires [Sutton et al. 2008], and analyzed Twitter use during natural hazard events to understand public situational awareness [Starbird et al. 2010; Vieweg et al. 2010]. This research program explicitly framed citizens as active participants in emergency information ecosystems rather than passive recipients of official communications.

The 2010 Haiti earthquake catalyzed a major research thread on digital volunteerism that further cemented this orientation. Studies of “Voluntweeters” documented how remote volunteers relayed, verified, and structured crisis information through microblogging platforms [Starbird and Palen 2011, 2013]. Subsequent work examined humanitarian mapping through OpenStreetMap [Herfort et al. 2021], crowdsourcing platforms like AIDR that combine volunteer labeling with machine learning [Imran et al. 2015, 2014], and the formalization of volunteer networks into organizations like Humanity Road and Virtual Operations Support Teams [Fathi et al. 2020; Norris et al. 2022]. Research on misinformation and rumor during crises has similarly analyzed public Twitter data to understand collective sensemaking, rumor propagation, and correction dynamics [Arif et al. 2016; Krafft et al. 2017; Starbird et al. 2014, 2016]. While this body of work has generated valuable understanding of how the public and how volunteers navigate crisis information, the dominant methodology—social media data analysis—and primary participant populations have not EM perspectives.

Studies that include professional EMs as research participants remain notable exceptions in the crisis informatics literature. Amanda Hughes's sustained research thread on Public Information Officers represents the most direct engagement with professional EM social media practices through participatory design and interviews [Hughes and Palen [n. d.]; Hughes and Shah 2016; Hughes et al. 2014]. A small number of ethnographic studies have examined emergency dispatch centers and operations centers as sites of collaborative work [Cavaliere et al. 2020; Martin et al. 1997; Pettersson et al. 2002]. European research has examined first responder technology acceptance through multi-year ethnography with voluntary fire departments [Reuter et al. 2018; Weidinger et al. 2024]. Recent work has begun to address this gap more directly, with a survey of 126 U.S. EM personnel explicitly calling for crisis informatics researchers to develop technology for EMs [Toups Dugas and Semaan 2023], and emerging studies examining the coordination challenges at the intersection of professional responders and digital volunteers [Hughes and Tapia 2015]. CSCW literature on emergency coordination has documented significant challenges including multi-agency information sharing difficulties and real-time collaboration requirements [Bharosa et al. 2010; Mendonça et al. 2007], yet few studies conduct ethnographic observation or in-depth interviews to understand how professional EMs actually accomplish their technology-mediated work. This gap has practical significance for responsible AI development. Professional EMs make consequential decisions about technology adoption, AI tool use, and information practices under organizational, legal, and political constraints quite different from volunteer or public contexts [Moitra et al. 2022]. They coordinate official response within bureaucratic structures, maintain accountability for life-safety decisions, and navigate resource constraints that shape what technologies are feasible to adopt. Understanding their technology use cases and perspectives on AI—through qualitative research directly with practitioners—addresses a documented gap and ensures that insights informing future system development reflect the needs and constraints of those who will ultimately deploy these tools in operational settings.

C.2 Domain Expert Perspectives on AI in High-Stakes Contexts

A growing body of FAccT and HCI research examines how domain experts across professional sectors perceive, adopt, and critique AI systems deployed in their work [Yu et al. 2025]—but very few focus on emergency management. Healthcare has been the most extensively-studied domain, with qualitative investigations revealing that clinicians consistently emphasize the necessity of human oversight, express concerns about algorithmic opacity in complex cases, and worry about deskilling effects from AI reliance [Elgin and Elgin 2024; Funer et al. 2024; Nouis et al. 2025]. Yoon et al. found that when AI recommendations contradict professional judgment, practitioners typically consult colleagues before reconsidering—suggesting that human expertise networks remain central to high-stakes decision-making in AI-augmented environments [Yoon et al. 2022]. Research on clinical decision support systems has developed conceptual frameworks around epistemic trust, examining the conditions under which professionals consider AI outputs credible enough to act upon. Frameworks for appropriate trust distinguish over-reliance from under-reliance and identify strategies for calibrating human-AI collaboration [Cabitza et al. 2023; Lai et al. 2023]. Understanding the landscape of conditions is critical for developing emergency management where life-safety decisions must be made rapidly under uncertainty.

Research on the Allegheny Family Screening Tool (AFST) in child welfare has yielded sophisticated analyses of frontline workers interacting with algorithmic decision support. These studies demonstrate that workers' reliance on AI is shaped by their access to rich contextual information beyond what models capture, their beliefs about AI capabilities versus their own expertise, and organizational pressures around tool use [Kawakami et al. 2022a,b]. Mixed method analysis showed that human-algorithm collaboration reduced racial disparities in screening decisions compared to algorithm-only predictions [Cheng et al. 2022]. This evidence that human oversight can further refine algorithmic outputs has important implications for emergency management, where equity in resource allocation between communities is a long-standing priority [Federal Emergency Management Agency 2011]. Alkhatib and Bernstein introduced “street-level algorithms,” algorithmic systems that make on-the-ground decisions about humans and offer design recommendations to help mitigate harm caused by nonsensical decisions [Alkhatib and Bernstein 2019].

Research in criminal justice reveals that organizational context shapes AI tool adoption as much as individual attitudes. A study of judicial responses to a risk assessment instrument found that judges overwhelmingly ignored the tool, describing it as “useless” and “not helpful,” with non-use resulting primarily from county-level norms and information dissemination failures rather than individual distrust [Pruss 2023]. Cross-cutting theoretical work has examined deskilling—the degradation of professional competencies through reduced practice due to AI reliance—arguing that AI development frequently treats domain experts as data sources rather than essential partners, risking erosion of the very expertise systems purport to augment [Sambasivan and Veeraraghavan 2022].

While this literature provides rich methodological and theoretical foundations, emergency management remains notably absent from studies of domain expert perspectives on AI. One of the only recent studies centered on the needs of disaster risk practitioners included a population primarily of scientists and academics rather than operational EMs [Moitra et al. 2022]. Our study addresses this gap by centering the perspectives of frontline EMs who would ultimately use AI tools in operational settings, contributing to a more complete understanding of expert-AI interaction across high-stakes public sector contexts.

D Extended Methods

This appendix provides extended methodological detail supplementing Section 3. We describe our research questions, expanded recruitment and sampling procedures, pre-interview instrumentation, interview procedures, and complete sample characteristics.

D.1 Study Design Rationale

Our three-phase design served specific methodological purposes. The eligibility screening phase allowed us to verify participant qualifications before scheduling and informed purposive sampling decisions aimed at maximizing sample diversity. The pre-interview questionnaire reduced cognitive burden during interviews by collecting logistical and demographic information beforehand, while also providing contextual information on each participant's AI familiarity that we used to tailor interview questions. This phased approach enabled us to meet participants where they were rather than asking uninformed questions about AI experiences they may not have had.

D.2 Recruitment and Sampling Details

We recruited participants using purposive sampling with snowball recruitment [Patton 2015]. Our eligibility criteria required participants to be at least 18 years old and have a minimum of two years of professional experience in emergency management serving geographic areas within the United States. We initially contacted potential participants through the professional network of our EM coauthors and through LinkedIn. We also asked early participants to refer additional contacts.

During recruitment, we observed geographic clustering, with early participants disproportionately concentrated in one state and predominantly representing local public-sector roles—likely a function of our early participants' referrals. To increase sample diversity, we employed maximum variation sampling [Patton 2015] by requesting access to professional distribution lists in other states and jurisdictions. We explicitly sought variation across geographic regions, jurisdictional levels (local, county, state, federal), and employer types (public sector, private sector, nonprofit, academic).

We do not claim that our sample is representative of the broader EM population; rather, we sought sufficient diversity to surface a range of perspectives and experiences regarding technology and AI in emergency management.

D.3 Pre-Interview Instrumentation

D.3.1 Eligibility Screening Survey. The eligibility screening survey collected: contact information, age range, years of EM experience, current employer, current job title, and geographic area(s) served in their current EM jurisdiction. We used these responses to verify eligibility (age 18 or older and 2 or more years of US-based EM experience) and inform sampling decisions.

D.3.2 Pre-Interview Questionnaire. The pre-interview questionnaire collected scheduling availability and two categories of information:

- (1) **AI familiarity**, using questions adapted from Pew Research Center's survey on Americans' experiences with AI [Kennedy et al. 2023]:
 - "How much have you heard or read about AI?" [a lot, a little, nothing at all]
 - "How often do you interact with AI?" [very frequently, often, rarely, never]
 - "How often do you interact with AI as a part of your job in disaster management?" [very frequently, often, rarely, never]
- (2) **Optional demographics.** All questions included a "prefer not to answer" response and were formatted as follows:
 - multiple choice [disability status, education level, veteran status]
 - free response [gender, sexuality]
 - numerical Likert scale [social political orientation, economic political orientation]

Using validated items from a national survey enables potential comparison of our sample's AI familiarity to broader population data. Responses to AI familiarity questions informed how we tailored interview questions: for participants indicating AI use at work, we probed their specific applications in detail; for those indicating rare or no AI interaction, we focused on general technology experiences and hypothetical AI applications.

D.4 Interview Protocol Development

We developed our semi-structured interview protocol through an iterative, expert-informed design process. The first author drafted initial interview questions based on our research objectives and subsequently refined them with the following domain expert coauthors:

- **EM practitioner-faculty collaborators** (David Merrick and Robert McDaniel) reviewed the questions to ensure alignment with emergency management terminology and professional contexts, identifying potentially confusing vocabulary and suggesting domain-appropriate alternatives.
- **Field robotics researchers** (Dr. Robin Murphy and Thomas Manzini) with extensive EM experience provided feedback on the technical framing of AI-related questions.

Following these consultations, we submitted all materials to Carnegie Mellon's Institutional Review Board and received approval under exempt status (Category 2: Interactions involving surveys, interviews, and observation of public behavior).

D.4.1 Protocol Structure Rationale. The interview protocol employed deliberate design strategies to elicit rich, grounded responses:

Funnel approach: We structured questions moving from broad recall-based questions to specific evaluative questions [Morgan 1998]. Interviews began by asking participants to describe their job responsibilities, work experience, and disaster response history before introducing questions about technology or AI. This sequencing cognitively grounded participants in their concrete professional experiences before asking them to reflect or speculate.

Scenario-based elicitation: For the substantive portion, we employed the critical incident technique [Flanagan 1954]. Rather than abstract questions about technology use, we asked participants to recall and walk through specific decision-making scenarios they had encountered. We systematically probed for details including the decision process, information sources consulted, stakeholders involved, and outcomes. Only after participants narrated these concrete experiences did we ask evaluative questions about process improvements and potential technology contributions.

This approach—grounding evaluative questions in recalled experiences—helps participants anchor responses in actual practice rather than hypotheticals, yielding more ecologically valid insights into real-world technology needs and constraints.

D.5 Interview Procedures

The first author conducted all 32 semi-structured interviews via video conference (Zoom) between March 2025 and April 2025. Interviews lasted between 60 and 90 minutes, with the majority approximately 75 minutes, yielding approximately 41 hours of total audio.

At each interview's start, the interviewer verbally confirmed informed consent, reminded participants of audio recording, and answered study questions. All participants consented to audio recording. Recordings were captured using Otter.ai, an automated transcription service accessed through an institutional license. Video was not recorded to protect participant privacy; participants could optionally enable their own video.

The interviewer used a standardized set of core questions while allowing flexibility to probe responses, follow relevant tangents, and adapt question wording to conversation flow. When participants mentioned technologies

that earlier participants had also discussed, the interviewer asked follow-up questions to probe their relative commonality across the profession—an example of iterative, cross-case probing [Miles et al. 2014].

After each interview, the first author reviewed automated transcripts for accuracy, correcting transcription errors. Transcripts were de-identified by removing names, specific employer names, and other potentially identifying information. Each participant was assigned a unique identifier reflecting state and interview sequence (e.g., FL3, MI1).

D.6 Complete Sample Characteristics

Our final sample comprised 32 emergency management professionals (Table 2). Years of experience ranged from 5 to 45 years ($M = 20.7$, $Mdn = 21.5$). This senior, experienced sample reflects both snowball recruitment through professional networks and the reality that individuals with substantial work history are more likely to hold diverse perspectives on the profession's evolution.

The sample was highly educated: 50.0% held master's degrees, 28.1% held bachelor's degrees, 12.5% held doctoral degrees, and the remainder held associate degrees, professional degrees (e.g., JD), or had completed some college without a degree.

For political orientation, participants' median responses on 7-point scales (1 = extremely liberal, 7 = extremely conservative) were 4.0 for social issues ($M = 4.0$, $SD = 1.4$) and 5.0 for economic/fiscal issues ($M = 4.5$, $SD = 1.5$), indicating a moderate to slightly conservative-leaning sample.

All demographic items were optional; response rates ranged from 69% to 100%. Given our sample size and limited variation on several dimensions, we report demographics descriptively to characterize our sample rather than for subgroup analysis.

D.7 Data Analysis Details

We analyzed interview transcripts using reflexive thematic analysis following Braun and Clarke's six-phase approach [Braun and Clarke 2006, 2019]. This method is well-suited to exploratory qualitative research seeking to identify patterns of meaning across a dataset while maintaining transparency about the researchers' interpretive role.

The first author conducted an initial round of open coding in ATLAS.ti, generating 847 initial codes across the 32 transcripts. Codes captured both semantic content (e.g., "uses ChatGPT for message drafting") and latent meaning (e.g., "frustration with legacy systems").

The second and third authors then independently read a subset of transcripts, surfaced preliminary insights, and met with the first author to discuss emerging patterns. Through two additional rounds of iterative coding, the research team reviewed, combined, and synthesized related codes into candidate themes. We employed constant comparison techniques [Miles et al. 2014] to refine theme boundaries and ensure internal coherence.

This process yielded the thematic structure presented in the main text: contextual factors shaping technology adoption, participants' perceptions of AI, and seven functional use case categories comprising 136 distinct technology use cases.

D.8 Codebook

Table 3. Final codebook (v3.0) used in the reflexive thematic analysis of the $N = 32$ interview transcripts. Codes are organized under five top-level themes (each introduced by a bolded header row). Illustrative examples are paraphrased or composite excerpts representative of the kinds of statements coded at each node, not verbatim participant quotations.

Code	Definition and illustrative examples
Theme 1: Decision-Making Scenarios / Processes. <i>A participant's description of a specific incident in which they made a decision. To be coded under this theme, an excerpt must cover: (1) the decision-making process, (2) the factors or rationale behind the decision, and (3) at least one reference to a historical occurrence (something that actually happened at least once). This theme is distinct from a description of job duties, because it refers to a scenario that actually occurred. E.g., "Do we stand up the EOC?" followed by specific details.</i>	
Description of scenario & context of decision	Exposition about the detailed scenario and the context for a decision-making process, along with the process itself.
Factors of decision	Constraints, features, factors, or reasons considered in making a choice. E.g., "You have to evacuate Galveston before you evacuate Friendswood, Houston, etc."; "There are four things that affect every decision: liability, the media, funding, and politics."
Limitations of / opportunities to improve the DM process	<i>Opportunities:</i> things that could be added or improved to help make decisions. <i>Limitations:</i> things already present in the process that hinder decision-making.
Theme 2: Technology Use Cases for EM. <i>Current and/or imagined use cases for technology in an EM context. This code group is a cross-cutting subset of the codes Current Tech Usage → Current AI use, Current Tech Usage → Existing Usage Scenario, and Future Tech Imaginings → Opportunities. The seven Level-1 codes below correspond to the seven use case categories reported in Section 5 and Appendix B.</i>	
Public & stakeholder messaging	Coordinated, two-way communication delivering timely, accurate, and consistent information to the public and to relevant stakeholders before, during, and after a crisis. E.g., operational coordination with EM partners; emergency messaging and IPAWS alerts; preparedness, response, and recovery information dissemination to the public.
Situational awareness	The ability to perceive, understand, and project the meaning of elements in the operating environment to inform decision-making and action. Involves continuously gathering and analyzing data, anticipating developments, and using that understanding to assess risks, coordinate response efforts, and allocate resources. E.g., combining disparate data streams, filtering irrelevant data, and generating insights from large volumes of data.
Planning	A systematic, ongoing process to develop and maintain comprehensive strategies for preventing, preparing for, responding to, and recovering from crises. E.g., plan development (Incident Action Plan; Continuity of Operations Plan, Business Continuity Plan, or Comprehensive Emergency Management Plan); plan testing and evaluation ("crosswalk").
Damage assessment	The systematic process of evaluating the nature and extent of harm and losses to a community's property, infrastructure, and population following a disaster. Primary goals are to (1) inform and guide the allocation of resources for response and recovery and (2) determine eligibility for, and the need to request, state or federal disaster assistance. E.g., rapid needs assessment; building damage assessment.
Incident management	Determining the level of activation / scale of response; references and checklists used during incident response; tracking and coordinating resource usage.

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Code	Definition and illustrative examples
Search & rescue	Operations during the response phase of EM to locate missing persons and remove them from hazards. E.g., identifying objects of interest from drone footage (live or pre-recorded) to assist in a search (e.g., “the missing person is wearing a blue shirt”).
Exercises & training	Activities for training personnel and validating plans in a non-operational setting. E.g., exercise scenario creation and inject / prompt augmentation; exercise design; after-action report creation; exercise evaluation.
Theme 3: EM Context. <i>EM-related tasks, capabilities, and objectives mentioned by participants, regardless of whether they currently involve technology. Includes job functions and responsibilities, disaster types, descriptions of disaster phenomena, and other contextual factors that shape EM work.</i>	
Non-tech-related limitations & opportunities	External (non-technological) factors that help or hinder EMs in completing their job functions. Includes participants’ strategies for and opinions on how to address these limitations and opportunities. E.g., “Public officials don’t know what EM is”; chronic under-funding; “the public doesn’t read or understand alerts.”
Disaster types	Incident types EMs have encountered. E.g., flooding; a planned event; an unplanned event.
EM philosophy, rules of thumb	High-level descriptions of priorities or heuristics that EMs use to execute or inform their job duties. E.g., “All disasters are local”; “Top priorities in response are: (1) life safety, (2) property, (3) the environment.”
Relevant laws, acts, & regulations	Formal government-enforced rules (laws, regulations, etc.) that affect EM work. Also yields, or can be operationalized via, standardized formal processes. E.g., ICS, NIMS, the EOC structure; FEMA Incident Action Plan forms; FEMA Individual Assistance and Public Assistance programs.
Job categories, roles, & duties	Details about an EM-related role (full-time or temporary) that the participant has held or interacted with, and that role’s associated duties or functions. May map to incident-response positions or to full-time jobs, and may include certifications (e.g., MEP). E.g., intelligence officer, planning chief, GIS specialist, PIO, incident commander; “I am planning chief, and I interact with the intelligence officer, who does ___.”
Specific job setting & background	Details specific to a participant’s particular job context: external factors important to understanding the execution of their job duties that they (the EM) cannot change. May include opinions. Not generalizable across EMs. E.g., “Our county is generally very rural, but recurring political events bring in tens of thousands of visitors”; “There are many Urdu and Vietnamese speakers in our area”; “Flooding is the primary natural hazard for our region”; “We have seven hospitals across the Miami area”; “Our county has many seasonal visitors during the summer”; “We receive grant funding that requires hosting tabletops every six months.”
Theme 4: Current Tech Usage. <i>Existing software, hardware, and/or IoT products and features—including AI—that participants currently use or have used in an EM context. Annotations include both off-hand mentions of tools and full descriptions of how a tool is used in practice.</i>	
Existing products	Names of specific tech tools used.
Existing usage scenario	The context in which a participant currently uses, or previously used, an existing technology.
Current AI use	A real instance of using AI to do something, whether job-related or not.

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Table 3 continued from previous page.

Code	Definition and illustrative examples
Limitations of existing tech	Must be with respect to a specific existing product. Includes both poorly implemented features (“the feature exists but is limited”) and features that are absent from existing products.
Successes	Features that work, features participants like, and positive experiences with existing tech. E.g., “X is great for ___”; “X helps me ___ quickly”; “I really like ___ about X.”
Theme 5: Future Tech Imaginings. <i>How participants imagine technology (including AI) may or may not be useful in EM contexts but is not currently in their workflow (and may not yet exist). Includes feature suggestions for existing software as well as fully speculative tools.</i>	
Opportunities	Forward-looking suggestions about new features, devices, or software; novel applications of existing devices or software; or future usage scenarios that participants believe would be valuable. E.g., features absent from existing tech; products serving a function that does not yet exist; “AI would be good for ___.”
Concerns, guidance, & limitations (w.r.t. a specific opportunity)	Reservations or design guidance that a participant attaches to a specific imagined opportunity. E.g., “I want AI to do situational awareness, but I worry that it might ___”; “If we do building damage assessment with aerial footage, we lose 3D signals like waterlines.”
Opinions about AI	Participants’ high-level views on AI’s capabilities and usefulness. E.g., “AI cannot replace human EMs”; “AI can be a force multiplier for EMs”; “AI is really good at generating ideas for public messaging.”

D.9 Summary Tables

Table 4. Technology Ecosystem Reported by Emergency Managers. Tools participants described using, organized by functional category.

Category	Tools
Communication	WebEOC, Microsoft Teams, Slack, RAVE Alert, Everbridge
Mapping/GIS	ArcGIS, Google Earth Pro
Damage Assessment	Skydio drones, Fusus integration platforms
Planning	Microsoft Office suite, SharePoint
Social Media	Facebook, Twitter/X (manual monitoring)

Table 6. EM Technology Use Cases Summary

Use Case Category	Count	AI Used?
Public Messaging	24	Yes (ChatGPT)
Situational Awareness	15	No
Planning	12	No
Incident Management	11	No
Damage Assessment	10	Limited
Exercises	8	Yes (ChatGPT)
Search and Rescue	6	No

Table 7. Cross-Cutting Themes Across All Use Cases

Theme		Description
Information	Over-load	Fundamental challenge across all use cases; lack of synthesis tools
Flexibility	Reqs.	Frequent role changes and varying incident types require adaptable systems
Pragmatic	Innovation	Willingness to adopt “good enough” solutions due to resource constraints

E Interview Script

Background

Let's start with your work background.

- (1) In a few sentences, can you please describe your work experience related to emergency management?
- (2) Roughly how many disaster response efforts have you participated in?
- (3) What types of disaster events have you responded to?
- (4) How would you describe your current position and job responsibilities?
- (5) *[If not answered in 4]* What is your typical role in disaster response efforts?
- (6) *[If indicated in their form response that they use AI in their job]* Can you please describe how you use AI in your job?

Decision-making Scenarios

Now that I have a better understanding of your job functions, I'd love to learn more about specific decisions you make in the context of your job.

- (7) Can you please describe what you consider to be a critical decision in emergency management that you have been a part of?
 - a. In that scenario, can you please tell me about the decision process?
 - b. What was your role in the process?
 - c. What data or factors did you consider when making your decision?
 - d. What information were you given during that scenario?
 - e. What did you know about the situation before you were presented with this decision scenario?
 - f. What did you know about the stakeholders (people who will receive the resource) before you were presented with this decision scenario?
 - g. Did you decide in a team or on your own?
 - h. Once you made a decision, can you run me through what happens next?
 - i. Who or what was responsible for the execution of the next steps?

Decision-making Scenarios with Resource Constraints

- (8) Have you ever encountered a situation where the number of requests for a resource exceeded the resources available?
 - a. In that scenario, can you please tell me about your decision process for allocating resources?
 - b. What data or factors did you consider when making your decision?
 - c. What information were you given during that scenario?
 - d. What did you know about the situation before you were presented with this decision scenario?
 - e. What did you know about the stakeholders (people who will receive the resource) before you were presented with this decision scenario?
 - f. Did you decide in a team or on your own?
 - g. Once you made a decision, can you run me through how the resources were allocated?
 - i. *[If not covered]* Who or what was responsible for the distribution of resources?
- (9) *[If time permits]* Have you encountered the same type of resource decision where *[add details based on their previous description]* in other disaster events?
 - a. Can you tell me more about the similarities between those situations?
 - b. Can you tell me more about the differences between those situations?

- (10) *[If time permits]* Have you encountered different situations with a different type of limited resources and high demand? Can you please describe them for me?

Protective Actions Decision-making Scenarios

- (11) Have you participated in making decisions about protective measures (protective actions) before a forecasted disaster?
- What type of protective measures were involved?
 - What population was affected?
 - In that scenario, can you please tell me about the decision process?
 - What data or factors did you consider when making your decision?
 - What information were you given during that scenario?
 - What did you know about the situation before you were presented with this decision scenario?
 - What did you know about the stakeholders (people who are affected by the protective measures) before you were presented with this decision scenario?
 - Did you decide in a team or on your own?
 - Once you made a decision, can you walk me through how the protective measures were executed?
 - [If not covered]* Who or what was responsible for executing the protective measures?
- (12) *[If not covered in previous question(s)]* Have you participated in making decisions about evacuation plans before a forecasted disaster?
- What type of evacuation plans were involved?
 - What population was involved?
 - In that scenario, can you please tell me about the decision process?
 - What data or factors did you consider when making your decision?
 - What information were you given during that scenario?
 - What did you know about the situation before you were presented with this decision scenario?
 - What did you know about the stakeholders (people who are affected by the protective measures) before you were presented with this decision scenario?
 - Did you make a decision in a team or on your own?
 - Once you made a decision, can you walk me through how the evacuation measures were executed?
 - [If not covered]* Who or what was responsible for executing the evacuation measures?
- (13) *[If not covered in previous question(s)]* Have you participated in making decisions about the pre-positioning of resources before a forecasted disaster?
- In that scenario, can you please tell me about your decision process as to how to decide where to pre-position resources?
 - [If not covered]* Did you make use of logistic staging areas?
 - Can you tell me more about logistic staging areas?
 - What data or factors did you consider when making your decision?
 - What information were you given during that scenario?
 - What did you know about the situation before you were presented with this decision scenario?
 - What did you know about the stakeholders (people who will receive the resource) before you were presented with this decision scenario?
 - Did you decide in a team or on your own?
 - Once you made a decision, can you run me through how the resources were pre-positioned?
 - Who or what was responsible for the pre-positioning of resources?

Opportunities to Improve Decision-making Scenarios

Thank you so much for your detailed responses! Now that we've discussed some interesting decision scenarios, I'm curious about your perspective on how things could be done better. Please answer the following questions keeping in mind the decision scenarios we talked about before.

- (14) In the scenarios we discussed before, how effective was the amount of information/general setup of the process for you in making your decision? Why or why not?
- (15) In any of the scenarios we discussed before, what do you think could help improve the decision making process?
- (16) Now, let's focus on technology—where do you think AI technology could help improve your decision making process?
- (17) *[If they mentioned using AI at job in pre-interview survey]* How would you improve the AI you already use in your job?

Disaster Recovery Center Planning

Now, we're going to shift our focus to disaster recovery. We're interested in physical locations set up after a natural disaster where survivors can learn about financial assistance programs and get help applying for assistance.

- (18) What term do you use to describe such locations?

If they ask for clarification about what we mean, mention these may be called Disaster Recovery Centers or Multi-agency resource centers (MARC)s

- (19) Have you been involved in planning or supporting the placement of these “disaster recovery centers”¹?
- (20) *[if 19 is yes]* How long after a typical natural disaster are you/your department involved in administering “disaster recovery centers”?
- (21) Have you been directly involved in planning/placement of “disaster recovery centers”?
- (22) What plans are there for placement of “disaster recovery centers” before a disaster?
- (23) What factors do you/your team consider when deciding locations of these “disaster recovery centers”?
- (24) To staff and/or maintain these “disaster recovery centers”, are there partnerships worked out with local organizations and businesses worked out beforehand? How about with larger nonprofits?
- (25) After a disaster, how is information about “disaster recovery centers” shared with survivors?
- (26) *[If not covered in previous question(s)]* Does this information sharing include providing information to survivors in person?

Conclusion

- Before we wrap up, do you have any final thoughts about what we discussed today that you would like to share or address?
- Do you have any questions for me?

Thank you so much for your time.

¹From now on, we use “disaster recovery centers” as a placeholder for the term they identified in their response to #18